

Subject Description Form

Subject Code	ITC3207T
Subject Title	Analytical Methods for Fashion Business and Technology
Credit Value	3
Level	3
Pre-requisite(s) / <Co-requisite> / (Exclusion)	Nil
Objectives	The subject provides the knowledge to analyse fashion business and technology problems quantitatively, develops practical skills in solving quantitative problems in fashion business and technology with up-to-date software packages, as well as cultivates abilities in data processing and troubleshooting in fashion business and technology research.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - apply the appropriate techniques to present, analyse and interpret the numerical data generated in surveys and experiments in the fashion business and technology; - competently use popular software packages to prepare the data file of survey and experimental results obtained from fashion business and technology, and generate appropriate output for the data analysis; - interpret and analyse the output of software packages for data generated in surveys and experiments in the fashion business and technology; - analyse the numerical data published in fashion business and technology literature; - develop critical and creative thinking in identifying and solving quantitative problems in the fashion business and technology research.
Subject Synopsis/ Indicative Syllabus	(I) Planning Fashion Business and Technology Research Data collection with different sampling methods: random, stratified, cluster, quota, and convenience. Determination of data sample size. Presentation of data obtained from fashion business and technology in a concise and informative manner using appropriate tables, graphs and numerical measures. Preparation of the data file for fashion survey and experimental results, and generation of graphical and tabular output of software packages, e.g. SPSS.

	(II) Conducting Fashion Business and Technology Research Reliability analysis of survey results from fashion business. Setup of appropriate hypotheses for fashion business and technology problems, application of comparing group means, linear regression and correlation in quantitative analysis of survey and experimental data – practical examples and interpretation of applications in fashion business and technology. Practical application of time series and quality control for fashion business and textiles. Preparation of the data file of survey and experimental results obtained from fashion business and technology, and generation and interpretation of appropriate output of software packages, e.g. SPSS. (III) Analysis of Fashion Business and Technology Research Grouping of similar objects (survey respondents, fashion products etc.) into clusters using Cluster Analysis – practical examples and interpretation of applications in fashion business and technology. Clustering algorithms: Hierarchical and K-means. Search for hidden dimensions from a large number of variables in fashion survey and technology experiments using Factor Analysis – practical examples and interpretation of applications in fashion business and technology. Redundancy and correlation matrix. Factor loadings and extraction, effect of factor rotation and factor naming. Preparation of the data file of fashion survey and experimental results, as well as generation and interpretation of appropriate output of software packages, e.g. SPSS.																																															
Teaching/Learning Methodology	Lectures will focus on the introduction and explanation of key concepts and applications of analytical methods for fashion business and technology. Studio classes will provide students with the opportunity to strengthen their understanding of knowledge taught in lectures, and more importantly, to gain hands-on operation of popular software packages for quantitative analysis of fashion business and technology problems. Activities will include student discussions of problem sets and case studies. A variety of assessment tools will be used, including assignments, tutorial reports, written and hand-on tests, and final examination.																																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>1. Assignment/ exercise</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Report/test</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="5"></td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	Continuous Assessment	50%	✓	✓	✓	✓	✓	1. Assignment/ exercise	30%	✓	✓	✓	✓	✓	2. Report/test	20%	✓	✓	✓	✓	✓	Examination	50%	✓	✓	✓	✓	✓	Total	100%					
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	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Coursework and final examination will be important in assessing the intended learning outcomes. Coursework may include assignment, tutorial exercise and report, and term test. The final examination will assess the overall understanding of concepts and knowledge of quantitative analysis methods/software and their applications in fashion business and technology. The materials submitted for this assessment must be the student's own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students' own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.	
Student Study Effort Expected	Class contact:	
	• Lecture	26 Hrs.
	• Studio	12 Hrs.
	Other student study effort:	
	• Self-study	67 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	<u>Books</u> Aldrich, J. O., Cunningham, J. B. (2016), <i>Using IBM SPSS Statistics: An Interactive Hands-on Approach</i>. Sage Publications Inc, California. Darren, G., Paul, M. (2016), <i>IBM SPSS Statistics 23 Step by Step: a Simple Guide and Reference</i>. Routledge, New York. Everitt, B. S., Landau, S., Leese, M., Stahl, D. (2011), <i>Cluster Analysis</i>. John Wiley. Hair, J. F., Black, W. C., Babin, B. J., Anderson, R.E., Tatham, R.L. (2006), <i>Multivariate Data Analysis</i>. Pearson; Prentice Hall, NJ. Heeringa, S. G., West, B. T., Berglund, P. A. (2010), <i>Applied Survey Data Analysis</i>. Chapman & Hall/CRC. Kline, P. (1994), <i>An Easy Guide to Factor Analysis</i>. Routledge, New York. Pallant, J. (2016), <i>SPSS Survival Manual</i>. McGraw-Hill Education, England. Siegel, A. F. (2016), <i>Practical Business Statistics</i>. Academic Press, London. Spencer, N. H. (2014), <i>Essentials of Multivariate Data Analysis</i>. CRC Press.	